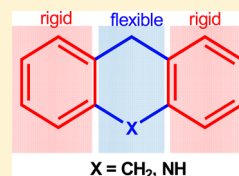


Isostructural Crystallization Behavior of Dihydroanthracene and Dihydroacridine

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S Supporting Information

ABSTRACT: The crystal packing motifs of acenes are usually built up by herringbone C–H... π interactions; on the contrary, *N*-heteroacenes form C–H...N contacts. In the case of dihydroazaacenes the molecular structure of a rigid aromatic system is interrupted by a flexible part. The influence of heterocyclic CH₂/NH substitution on the crystallization behavior and crystal packing of 9,10-dihydroacridine (DHAC) and 9,10-dihydroanthracene (DHAN) as well as a 1:1 mixture of both compounds was investigated by differential scanning calorimetry (DSC) measurements, hot stage microscopy methods, and crystal structure determination.



■ INTRODUCTION

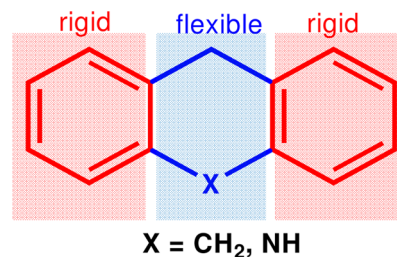
N-Heteroacenes are claimed to have great potential as organic semiconductors or for thin film transistor applications. Concerning their acene counterparts, they offer a higher stability and different frontier molecular orbital energies.^{1–3}

The molecular organization in the solid state of such compounds is usually dominated by herringbone packing motifs, which can be explained by the prevailing C–H... π interactions.^{4,5} In contrast, *N*-heteroacenes form C–H...N contacts and additionally organize themselves by offset π -stacking interactions.^{3,6}

Maly pointed out that, within polycyclic aromatic hydrocarbons, the replacement of CH units with nitrogen atoms leads to a significant impact on the solid state organization.⁶ The investigations on anthracene⁷ and the *N*-substituted analogues acridine^{8–13} and phenacine^{14–16} show that the crystal structures of the *N*-heterocycles are dominated by C–H...N dimer interactions while anthracene shows herringbone packing with C–H... π interactions. To explore the influence of the introduction of heteroatoms to a system on the crystallization behavior one has to differentiate between a fully aromatic ring system and an interrupted one. Hydrogenation of anthracene and acridine result in the formation of 9,10-dihydroanthracene (DHAN) and 9,10-dihydroacridine (DHAC). The most distinctive feature in the molecular structure of these both compounds is the bridging flexible unit between two rigid aromatic rings. There are few indications that compounds with such molecular structure could have comparable crystal packing properties.^{17–19} The phenomena of comparable and isostructural crystal packing has been investigated for substances where chlorine atoms were exchanged with CH₃ groups.²⁰ In these studies, the isostructural crystal packing is attributed to the similarity postulate of comparable volumes of the substituents. In addition, those investigations showed that the isostructural compounds formed solid solutions in all proportions.^{20,21}

To explore the limits of this similarity postulate, the influence of heterocyclic CH₂/NH substitution on the crystallization

behavior and crystal packing of dihydroacridine (DHAC), dihydroanthracene (DHAN), and a 1:1 mixture of both compounds was investigated by differential scanning calorimetry (DSC) measurements, hot stage microscopy methods, and crystal structure determination. For reasons of availability of chemicals, the deuterated *d*₃-9,10-dihydroacridine has been synthesized, while 9,10-dihydroanthracene was purchased by Fluka.



X = CH₂, dihydroanthracene (DHAN)

X = NH, dihydroacridine (DHAC)

■ EXPERIMENTAL SECTION

X-ray Crystallography. Single-crystal X-ray diffraction measurements of *d*₃-9,10-dihydroacridine and the cocrystal *d*₃-9,10-dihydroacridine/dihydroanthracene were carried out on a Rigaku XtaLab Mini diffractometer using graphite-monochromated Mo K α radiation ($\alpha = 0.71073$). Structures were solved by the direct method, and all non-hydrogen atoms were refined anisotropically on *F*². Refinement of the bridging CH₂/NH groups in the cocrystal *d*₃-9,10-dihydroacridine/dihydroanthracene had the occupational factors 0.7 for C and 0.3 for N (program SHELXTL-97, G.M. Sheldrick, University of Göttingen, Göttingen, Germany).

DSC Measurements. DSC of DHAC, DHAN (from Fluka, 97%), and 1:1, 1:2, 1:3, 2:1, 3:1 mixtures were carried out on a Netzsch DSC 204 Phoenix under nitrogen atmosphere using samples that were

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hermetically sealed in Al pans and heated. The samples were treated at a cooling and heating rate of 5 K min^{-1} . The melting points (T_M) were determined from the DSC thermograms during the programmed reheating steps, based on the onset temperatures.

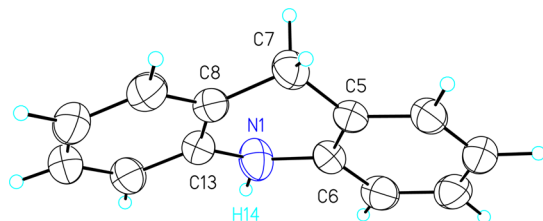


Figure 1. Molecular structure of DHAC.

Synthesis of d_3 -9,10-DHAC. For reasons of availability of chemicals, the deuterated d_3 -9,10-dihydroacridine has been synthesized and characterized. 9-Bromacridin (0.23 g, 0.89 mmol) with 32.4 mg of Pd/C (10%), 0.19 g of K_2CO_3 , and 16 mL of d_4 -methanol were placed in a two-necked Schlenk-flask. Sodium (0.51 g, 22.1 mmol) was put into a large test tube. The flask and the test tube were both equipped with a septum and connected by a cannula. A balloon was fixed to the second neck of the flask. The apparatus was flushed for about 10 min with nitrogen gas. Then, 2.0 mL of D_2O were added to the sodium. The reaction mixture in the flask was stirred in deuterium atmosphere for 4 h at room temperature. After the reaction was completed, the catalyst was filtered off and the remaining solvent was evaporated. The residue was dissolved in dichloromethane and washed with H_2O . The organic phase was then dried with MgSO_4 and concentrated under

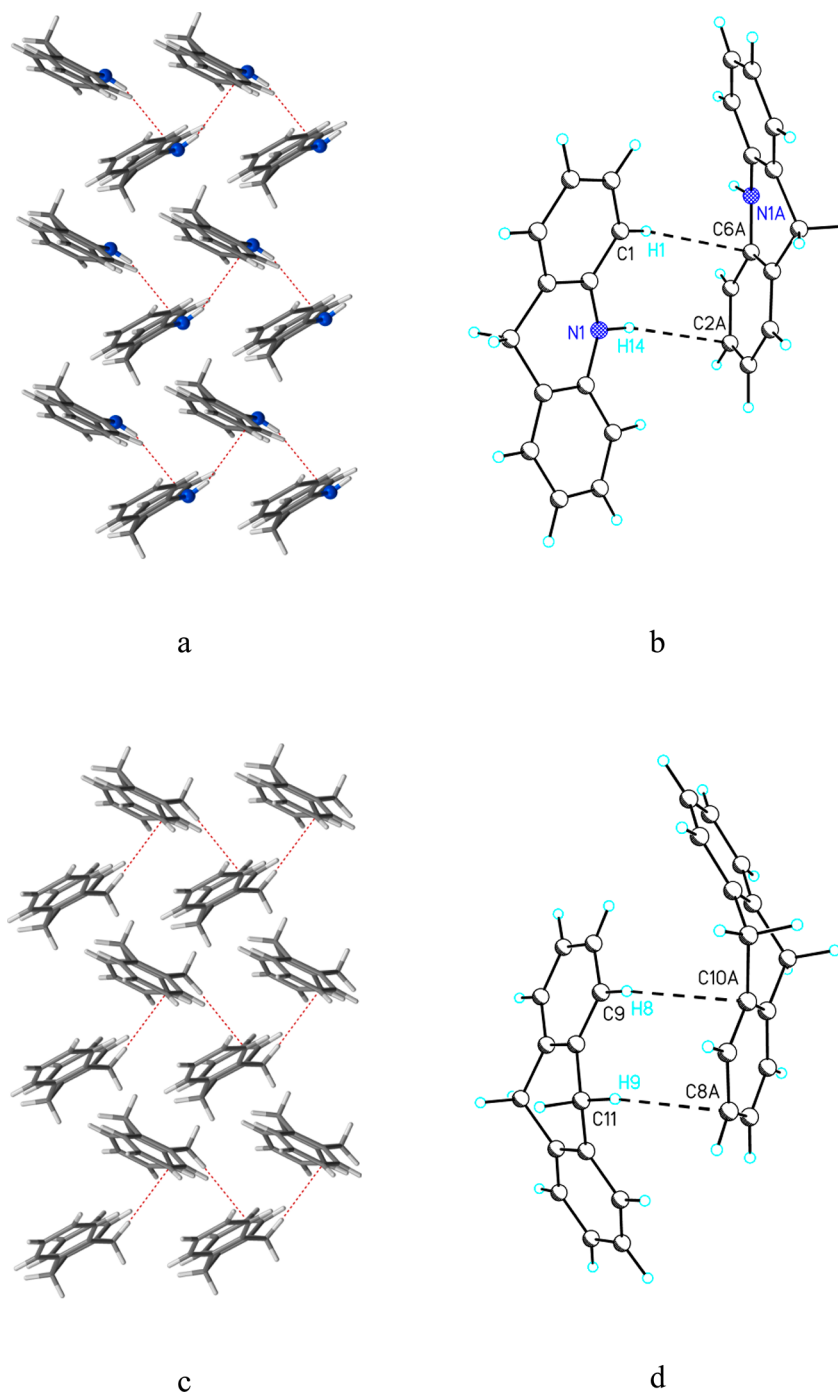


Figure 2. Crystal packing and $\text{NH}\cdots\pi$ -contacts/ $\text{CH}\cdots\pi$ -contacts of DHAC (a,b) and DHAN (c,d).

vacuum to give pale yellow crystals. Yield: 94 mg (0.50 mmol, 56.7%). ^1H NMR [200 MHz, CD_2Cl_2]: δ (ppm) = 7.12 (t, 4 H, J = 7.5 Hz), 6.84 (t, 2 H, J = 7.4 Hz), 6.67 (d, 2 H, J = 8.0 Hz). ^{13}C NMR [50 MHz, CD_2Cl_2]: δ (ppm) = 140.6, 129.1, 127.6, 121.1, 120.5, 113.9. EI MS [m/z]: 179.1, 180.1, 181.1, 182.1, 183.1, 184.1, 185.1. IR (ν [cm^{-1}]): 3375, 3031, 2965, 2927, 2850, 2110, 2075, 1919, 1603, 1577, 1450, 1415, 1307, 1287, 1250, 1182, 1154, 1119, 1094, 1044, 959, 940, 892, 818, 753.

Preparation of Binary DHAC/DHAN Mixtures. Ten milligrams of each composition were prepared. DHAC and DHAN (1:1, 1:2, 1:3, 2:1, and 3:1) were dissolved in 2 mL of acetone. The solvent was removed in a slow evaporation process. A 1:0.43 mixture was characterized by X-ray diffraction. DSC measurements were carried out on all prepared compositions (1:1, 1:2, 1:3, 2:1, and 3:1).

RESULTS AND DISCUSSION

Crystallography. The X-ray structure analysis of DHAC (Figure 1) shows a slightly angled molecule with a dihedral angle of 152.7° between the two aromatic rings. Compared to DHAN the analogous dihedral angle in DHAN is slightly reduced to 144.6° . Responsible for this observation are the angles in the centered flexible unit of DHAC. The observed bond lengths of DHAC are in the expected single bond range.

Despite the differences in the atomic composition, DHAC and DHAN²² crystallize isostructurally in the monoclinic space group $P2_1$ with nearly equal cell constants.

The crystal packing of both compounds is dominated by weak $\text{N-H}\cdots\pi$ and $\text{C-H}\cdots\pi$ contacts with comparable angles (Figure 2). By the fact that such weak intermolecular interactions span much wider ranges of angles and distances than strong intermolecular interactions,²³ a detailed interpretation of these observed weak intermolecular interactions is not needed.

It is known from Kitaigorodskii's "geometrical approach"²⁴ that the formation of solid solutions in mixed crystals is substantially influenced by the similarity of molecular size and shape as well as similar crystal packing of the components.^{15,24–29} As it has been shown that DHAC and DHAN exhibit comparable molecular skeletons and isostructural crystallization behavior, it seemed appealing to determine and analyze the crystal structure of a DHAC/DHAN mixture. The received X-ray analysis of the crystalline DHAC/DHAN mixture confirmed the presence of a mixture (Table 1). Similar to DHAN and DHAC, the DHAC/DHAN mixture crystallizes isostructurally in the monoclinic space group $P2_1$ with $Z' = 1$ and nearly equal cell constants. The analysis of the angles within the bridging flexible unit

($\text{C-CH}_2(\text{NH})-\text{C}$, $115.4(3)^\circ$) corresponds to the averaged values within DHAC (C-NH-C , 118.9°) and DHAN ($\text{C-CH}_2-\text{C}$, 112.1°). A similar situation is observed in the case of the bond lengths within the mixture. The C–N bond distances within DHAC are observed with 1.413(2) and 1.418(2) Å, whereas DHAN shows distances between C and CH_2 of 1.505(8) and 1.510(7) Å. The corresponding observed bond lengths within the mixture are 1.4615(4) and 1.467(4) Å. The dihedral angle of 150.8° , between the two aromatic rings in the mixture, corresponds to the mean value for DHAN and DHAC.

Thermal Behavior. A solid solution occurs in approximately 10% of all binary organic systems. This phenomenon is attributed to the similarity postulate stating that the presumption of solid solution formation depends on the molecular structure similarity of the used compounds.^{24,30} To investigate the crystallization and melting behavior of DHAN/DHAC mixtures, DSC measurements and hot stage microscopy methods have been performed.

The comparison of the first heating and cooling curves of DHAN, DHAC, and a 1:1 mixture (Figure 3) shows the

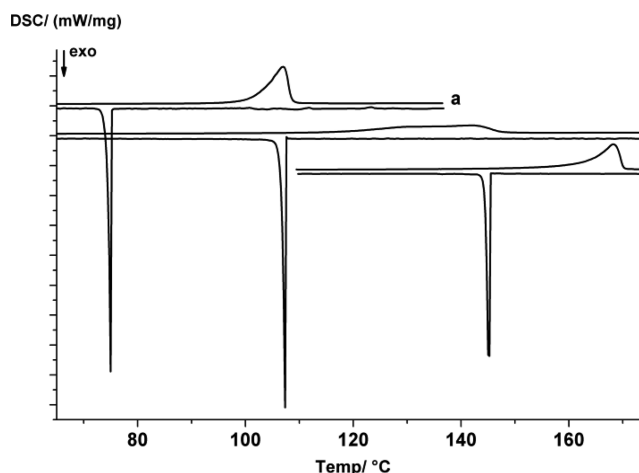


Figure 3. DSC diagrams of (a) DHAN, (b) DHAC/DHAN 1:1 mixture, and (c) DHAC. The figure shows one heating and one cooling cycle of each compound.

different melting and crystallization behavior of the investigated substances. DHAN melts at 101°C and crystallizes again at 74°C . The melting point of DHAC is located at 163°C , and the crystallization occurs at 145°C . The melting range of the solid solution is halfway between the two single compounds, starting from about 115°C and ending at 150°C . The onset temperature is calculated to be 118°C , while the highest peak happens to be at 142°C . The crystallization then occurs at 107°C and corresponds to the averaged values of DHAN and DHAC.

To distinguish whether the mixture of DHAC and DHAN forms a cocrystal or a solid solution, the system was analyzed by hot stage microscopy. A contact preparation was prepared by melting DHAC and DHAN by use of the Kofler hot bench.³¹ The transition region of the contact preparation of DHAN and DHAC has been examined for its thermal behavior (Figure 4). First, it should be noted that the two pure substances differ clearly in terms of their optical appearance. On the left side in the first image a turquoise surface is seen, which is interrupted by brown subjects, while on the right side crystalline needles with different colors (mainly pink) can be identified. The crystalline

Table 1. Crystallographic Data of DHAC, DHAN, and DHAC/DHAN Mixture

	DHAC	DHAN ²²	DHAC/DHAN
crystal system	monoclinic	monoclinic	monoclinic
space group	$P2_1$	$P2_1$	$P2_1$
a (Å)	7.897(4)	7.733(3)	7.75(2)
b (Å)	5.861(2)	6.245(3)	6.05(1)
c (Å)	10.737(5)	10.815(4)	10.71(2)
β (deg)	108.062(19)	108.34(5)	107.7(2)
V (Å ³)	472.5(4)	495.755	478.8(17)
Z	2	2	2
Z'	1	1	1
T (K)	173(2)	295	173(2)
reflections collected	4007	712	4075
independent reflections	1642	637	1689
parameter	171	174	177
R_1 [$I > 2\sigma_1$]	0.0485	0.045	0.0503

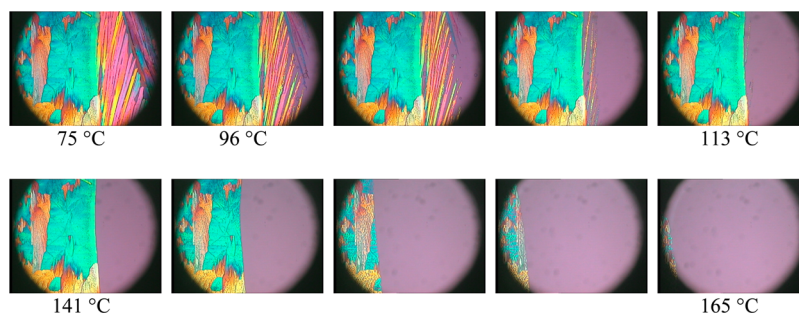


Figure 4. Hot stage micrographs to study the melting behavior of DHAC (left, turquoise and brown) + DHAN (right, pink), each crystallized from the melt.

needles on the right side represent DHAN, which begins to melt at 96 °C, while the left side with DHAC remains stable until about 141 °C then the crystalline film starts to melt slowly from right to left. At 165 °C only liquid is present. The behavior of a continuous melting process indicates the formation of a solid solution within the transition region of the contact preparation from the starting materials.

DSC diagrams as well as the generated phase diagram, in the Supporting Information of this article, reveal a continuous shift of the melting points within different proportions of the DHAC and DHAN mixture. This observation is consistent with the results of our HSM experiments.

CONCLUSIONS

Regarding isostructural crystallization behavior, the chloro-methyl exchange rule postulated by Desiraju and Sarma has successfully been transferred to the exchange of CH₂/NH in the case of 9,10-dihydroanthracene and 9,10-dihydroacridine.

From the presented data it is obvious that the examined compounds DHAN and DHAC crystallize isostructurally. While the non-hydrogenated acridine and anthracene compounds crystallize in different structures and are dominated by very different interactions, DHAC and DHAN form the same building blocks despite the CH₂/NH substitution and changes in the molecular geometry at the central bridge atom.

It has also been shown that mixtures of DHAC and DHAN form solid solutions in all proportions that are isostructural to the single compounds. Thus, Kitaigorodskii's similarity postulate of organic substances in a solid solution is also applicable to the binary system of DHAN and DHAC and should be considered in future investigations of related systems.

ASSOCIATED CONTENT

Supporting Information

Hot stage micrographs (HSM) of DHAN, HSM of DHAC, DSC diagrams of 1:1 solid solution and 1:1 physical mixture, DSC diagram of different compositions of DHAN and DHAC, phase diagram of the DHAC–DHAN crystallization system, and phase diagram of the DHAC–DHAN crystallization system + HSM of different compositions of DHAN and DHAC. CIF-files giving X-ray data with details of refinement procedures for DHAC and DHAC/DHAN 1:1 mixture. CCDC Nr. 986515 and 986552. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) Bunz, U. H. F. *Chem.—Eur. J.* **2009**, *15*, 6780–6789.
- (2) Miao, S.; Appleton, A. L.; Berger, N.; Barlow, S.; Marder, S. R.; Hardcastle, K. I.; Bunz, U. H. F. *Chem.—Eur. J.* **2009**, *15*, 4990–4993.
- (3) Winkler, M.; Houk, K. N. *J. Am. Chem. Soc.* **2007**, *129*, 1805–1815.
- (4) Nishio, M. *CrystEngComm* **2004**, *6*, 130–158.
- (5) Tsuzuki, S.; Fujii, A. *Phys. Chem. Chem. Phys.* **2008**, *10*, 2584–2594.
- (6) Maly, K. E. *Cryst. Growth Des.* **2012**, *11*, S628–S633.
- (7) Mason, R. *Acta Crystallogr.* **1964**, *17*, 547–555.
- (8) Kupka, A.; Vasylyeva, V.; Hofmann, D. W. M.; Yussenko, K. V.; Merz, K. *Cryst. Growth Des.* **2012**, *12*, S966–S971.
- (9) Lowde, R. D.; Phillips, D. C.; Wood, R. G. *Acta Crystallogr.* **1953**, *6*, 553–556.
- (10) Mei, X.; Wolf, C. *Cryst. Growth Des.* **2004**, *4*, 1099–1103.
- (11) Musumeci, D.; Hunter, C. A.; McCabe, J. F. *Cryst. Growth Des.* **2010**, *10*, 1661–1664.
- (12) Phillips, D. C. *Acta Crystallogr.* **1956**, *9*, 237–250.
- (13) Phillips, D. C.; Ahmed, F. R.; Barnes, W. H. *Acta Crystallogr.* **1960**, *13*, 365–377.
- (14) Herstein, F. H.; Schmidt, G. M. J. *Acta Crystallogr.* **1955**, *8*, 399–405.
- (15) Thalladi, V. R.; Smolka, T.; Gehrke, A.; Boese, R.; Sustmann, R. *New J. Chem.* **2000**, *24*, 143–147.
- (16) Wozniak, K.; Kariuki, B.; Jones, W. *Acta Crystallogr., Sect. C: Cryst. Struct. Commun.* **1991**, *C47*, 1113–1114.
- (17) Gerkin, R. E.; Lundstedt, A. P.; Reppart, W. J. *Acta Crystallogr., Sect. C: Cryst. Struct. Commun.* **1984**, *C40*, 1892–1894.
- (18) Gerkin, R. E.; Reppart, W. J. *Acta Crystallogr., Sect. C: Cryst. Struct. Commun.* **1986**, *C42*, 480–482.
- (19) Reppart, W. J.; Gallucci, J. C.; Lundstedt, A. P.; Gerkin, R. E. *Acta Crystallogr., Sect. C: Cryst. Struct. Commun.* **1984**, *C40*, 1572–1576.
- (20) (a) Desiraju, G. R.; Sarma, J. A. R. P. *Proc.—Indian Acad. Sci., Chem. Sci.* **1986**, *96*, 599–605. (b) Emery, P. R.; Thalladi, V. R. *Angew. Chem.* **2007**, *119*, 4210. (c) Bhattacharya, S.; Saha, B. K. *Cryst. Growth Des.* **2012**, *12*, 169.
- (21) Wei, Y.; Zhang, X.; Zhang, J.; Dang, L.; Wei, H. *Fluid Phase Equilib.* **2012**, *319*, 23–29.

- (22) (a) Herbstein, F. H.; Kapon, M.; Reisner, G. M. *Acta Crystallogr.* **1986**, *B42*, 181–187. (b) Reboul, P. J. P.; Oddon, Y.; Caranoni, C.; Soyfer, J. C.; Barbe, J.; Pepe, G. *Acta Crystallogr., Sect. C: Cryst. Struct. Commun.* **1987**, *C43*, 537–539.
- (23) Desiraju, G.; Steiner, T. *The Weak Hydrogen Bond: Applications to Structural Chemistry and Biology*; Oxford University Press: Oxford, U.K., 1999.
- (24) Kitaigorodsky, A. I. *Springer Series in Solid-State Sciences*; Springer: New York, 1984; Vol. 33 Solid Solutions.
- (25) Desiraju, G. R.; Vittal, J. J.; Ramanan, A. *Crystal Engineering: A Textbook*; World Scientific Pub Co: Singapore, 2011.
- (26) Fu, J.; Suuberg, E. M. *Fluid Phase Equilib.* **2013**, *342*, 60–70.
- (27) Kalman, A.; Parkanyi, L.; Argay, G. *Acta Crystallogr., Sect. B: Struct. Sci.* **1993**, *B49*, 1039–1049.
- (28) Oliveira, M. A.; Peterson, M. L.; Klein, D. *Cryst. Growth Des.* **2008**, *8*, 4487–4493.
- (29) (a) Ventola, L.; Calvet, T.; Cuevas-Diarte, M. A.; Ramirez, M.; Oonk, H. A. J.; Mondieig, D.; Negrier, P. *Phys. Chem. Chem. Phys.* **2004**, *6*, 1786–1791. (b) Frank, H. *Crystalline Molecular Complexes and Compounds*; Oxford University Press: New York, 2006; Vols. 1&2.
- (30) Kuduva, S. S.; Blaser, D.; Boese, R.; Desiraju, G. R. *Struct. Chem.* **2001**, *12*, 259–266.
- (31) Zencirci, N.; Gelbrich, T.; Kahlenberg, V.; Griesser, U. J. *Cryst. Growth Des.* **2009**, *9*, 3444–3456.